PALM OIL PLANTATION FEASIBILITY STUDY

Technical Analysis Report
Nigeria Palm Oil Profitability Model

Date: November 09, 2025
Prepared for: Nigerian-German Investor
Classification: Confidential - Investment Analysis

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY

This technical feasibility study provides a comprehensive analysis of palm oil plantation profitability in Nigeria, specifically addressing the question: At what scale and in which states will a plantation generate ■1 billion in annual profit? **Key Findings:**

- **Base Scenario:** Requires 5,602 hectares with ■18.0 billion total investment, yielding 5.6% ROI
- **Optimistic Scenario:** Requires 1,339 hectares with ■4.4 billion investment, yielding 22.8% ROI
- **Top States:** Cross River, Ondo, and Edo offer the best profitability potential
- **Payback Period:** 7.4 years (Optimistic) to 21 years (Base), including maturation

The analysis reveals that achieving ■1 billion in annual profit is feasible but requires significant scale, optimal agro-climatic conditions, and operational excellence. Conservative scenarios show losses, emphasizing the importance of targeting best-in-class performance metrics.

2. METHODOLOGY AND DATA SOURCES

2.1 Data Collection Approach

This study synthesizes data from multiple authoritative sources to build a comprehensive profitability model: **Primary Sources:**

• Okomu Oil PLC Annual Reports (2022-2024): Provided realistic yield benchmarks of 19.5 MT FFB/ha and cost structures from Nigeria's leading producer
• Presco PLC Annual Reports (2022-2024): Validated financial metrics with 18.2 MT FFB/ha yields
• FAO/FAOSTAT Production Database: National production statistics and historical trends

Supporting Sources:

• PIND Foundation palm oil sector mapping for regional analysis
• World Bank Climate Knowledge Portal for rainfall and temperature data
• Nigerian Institute for Oil Palm Research (NIFOR) for technical parameters
• Industry market reports for pricing trends and forecasts

2.2 Analytical Framework

Three scenarios were modeled (Conservative, Base, Optimistic) varying key parameters:

• Fresh Fruit Bunch (FFB) yields: 16-21 MT per hectare
• Oil Extraction Rate (OER): 19%-23%
• Crude Palm Oil (CPO) prices: ■480,000-560,000 per MT

State-level adjustments incorporated rainfall patterns, infrastructure quality, and land costs to provide location-specific profitability estimates.

3. NATIONAL PALM OIL INDUSTRY OVERVIEW

3.1 Production Trends

Nigeria's palm oil sector shows steady growth in production area and output, though average yields remain below optimal levels due to aging palm stands and suboptimal management practices.

Year	Production (MT)	Area (Ha)	Yield (MT/Ha)	Avg Price (■/MT)
2020	1,400,000	360,000	3.89	■180,000
2021	1,450,000	375,000	3.87	■250,000
2022	1,500,000	390,000	3.85	■320,000
2023	1,550,000	405,000	3.83	■450,000
2024	1,600,000	420,000	3.81	■520,000

4. COST STRUCTURE ANALYSIS

4.1 Annual Cost Breakdown

Operating a palm oil plantation involves significant capital and operating expenses. The analysis categorizes costs into capital investments (amortized over 25-year palm lifecycle) and annual operating expenses.

Scenario	Capital (Amortized)	Operating (Annual)	Total Annual Cost
Conservative	■54,000	■1,780,000	■1,834,000
Base	■47,200	■2,035,000	■2,082,200
Optimistic	■41,200	■2,240,000	■2,281,200

4.2 Cost Components

Major cost drivers include labor (maintenance and harvesting), fertilizers, processing, and transportation. The Optimistic scenario assumes higher costs due to more intensive management practices aimed at maximizing yields.

5. REVENUE MODEL

5.1 Revenue Streams
Palm oil plantations generate revenue from two primary products:
1. **Crude Palm Oil (CPO):** The main product, constituting 75-85% of revenue
2. **Palm Kernel:** A valuable by-product, contributing 15-25% of revenue

Revenue varies significantly based on Fresh Fruit Bunch (FFB) yields, Oil Extraction Rate (OER), and market prices. The table below shows projected revenue per hectare across scenarios.

Scenario	FFB Yield	CPO (MT/Ha)	Kernel (MT/Ha)	Total Revenue
Conservative	16.0	3.04	0.96	■1,632,000
Base	18.5	3.88	1.20	■2,260,700
Optimistic	21.0	4.83	1.47	■3,028,200

6. PROFITABILITY SCENARIOS

6.1 Scenario Analysis Results

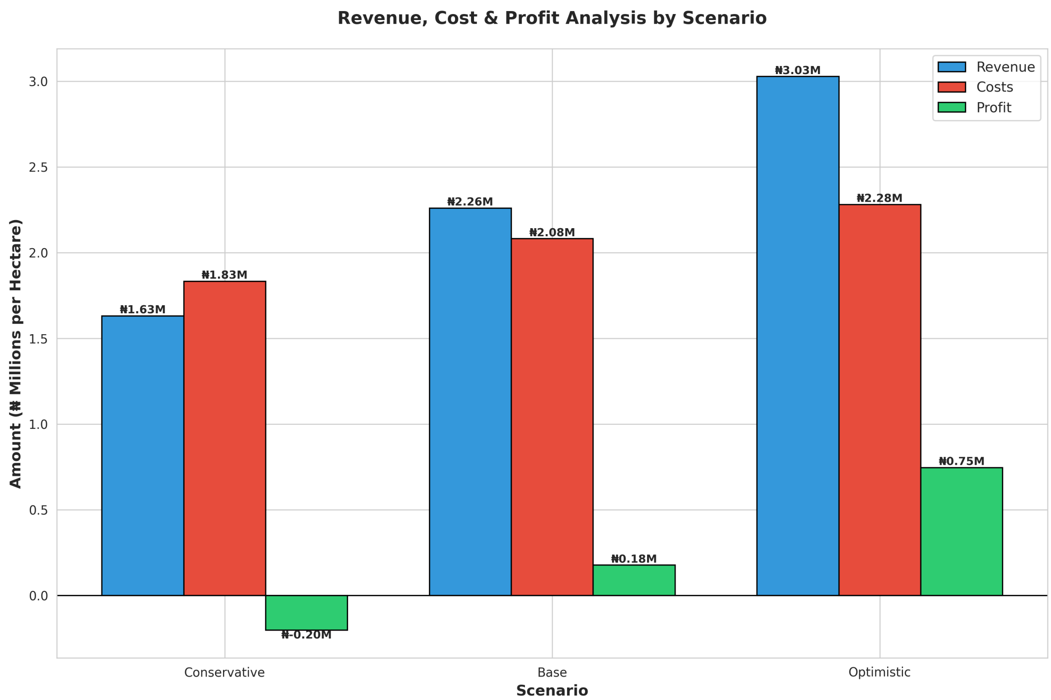
The profitability analysis reveals a wide range of outcomes depending on operational performance and market conditions. The Conservative scenario is not profitable, highlighting the critical importance of achieving competitive yields and efficient cost management.

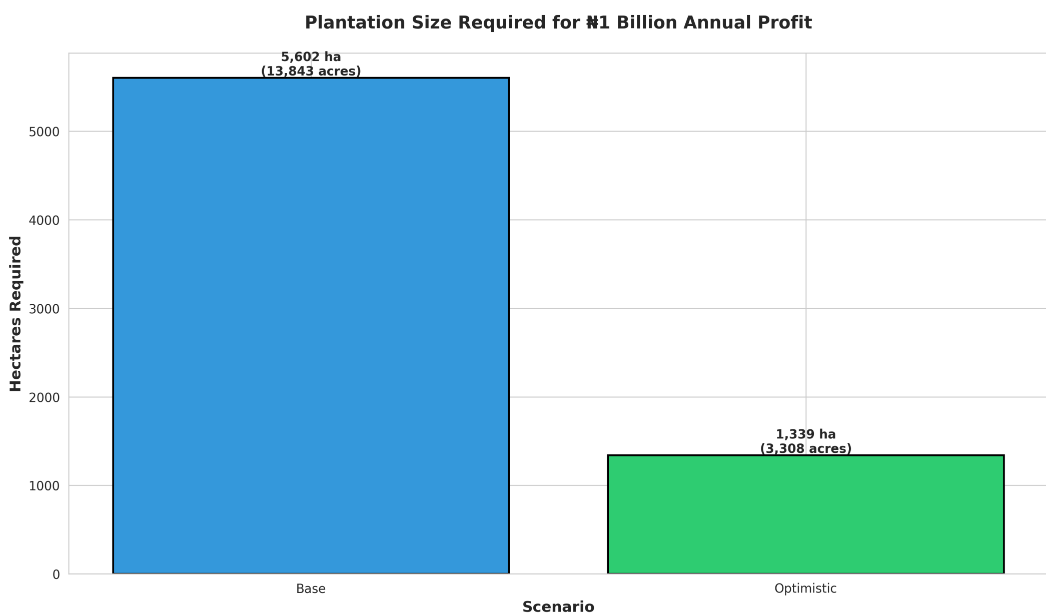
Scenario	Revenue/Ha	Cost/Ha	Profit/Ha	Margin (%)
Conservative	■1,632,000	■1,834,000	■-202,000	-12.4%
Base	■2,260,700	■2,082,200	■178,500	7.9%
Optimistic	■3,028,200	■2,281,200	■747,000	24.7%

6.2 Scale Requirements for ■1 Billion Annual Profit

To achieve the target of ■1 billion in annual profit, significant scale is required. The Base scenario requires 5,602 hectares (approximately 13,843 acres), while the Optimistic scenario reduces this requirement to 1,339 hectares (3,308 acres). This dramatic difference underscores the value of operational excellence and favorable market conditions.

6.3 Visual Analysis





7. STATE-BY-STATE ANALYSIS

7.1 Regional Profitability Assessment

State-level profitability varies significantly based on agro-climatic conditions, infrastructure, and land costs. The top three states (Cross River, Ondo, and Edo) show positive profitability, while other states face challenges that make them less attractive for immediate investment.

Rank	State	Profit/Ha	Rainfall (mm)	Infrastructure	Status
1	Cross River	■392,253	2800	7/10	Profitable
2	Ondo	■266,059	2200	7/10	Profitable
3	Edo	■173,776	2400	8/10	Profitable
4	Akwa Ibom	■-20,239	2600	6/10	Not Viable
5	Delta	■-146,433	2300	6/10	Not Viable
6	Imo	■-306,537	2400	5/10	Not Viable

8. INVESTMENT REQUIREMENTS

8.1 Capital Requirements Analysis

Achieving ■1 billion in annual profit requires substantial upfront investment covering land acquisition, plantation establishment, infrastructure, and working capital. The investment varies significantly based on the scale required under each scenario.

Scenario	Hectares	Capital Inv.	1st Yr Operating	Total Upfront
Base	5,602	■6.61B	■11.40B	■18.01B
Optimistic	1,339	■1.38B	■3.00B	■4.38B

9. ROI AND PAYBACK ANALYSIS

9.1 Financial Returns Assessment

Return on Investment (ROI) and payback periods vary significantly across scenarios. The Optimistic scenario offers attractive returns with 22.8% ROI and 7.4-year payback, while the Base scenario shows more modest but still viable returns. Payback periods include a 3-year maturation phase before palms reach full production.

Scenario	Total Investment	Annual Profit	ROI (%)	Payback (Years)
Base	■18.01B	■1.00B	5.6%	21.0
Optimistic	■4.38B	■1.00B	22.8%	7.4

10. RISK FACTORS AND ASSUMPTIONS

10.1 Key Assumptions

- Palm oil prices remain at ■480,000-560,000 per MT
- Oil extraction rates of 19%-23% are achievable
- FFB yields of 16-21 MT/ha can be maintained
- Land acquisition proceeds without major delays
- Rainfall patterns remain consistent with historical averages
- Labor remains available at projected costs
- No major pest or disease outbreaks
- Infrastructure improvements continue in target states

10.2 Risk Factors

Market Risks:

- Price volatility in palm oil markets could significantly impact revenues
- Competition from imported palm oil may affect local prices
- Currency fluctuations affecting input costs (especially imported equipment)

Operational Risks:

- Yield variability due to weather, pests, or disease
- Labor shortages or cost increases
- Equipment failures or maintenance issues
- Processing facility operational challenges

Regulatory and Political Risks:

- Changes in agricultural policies or taxation
- Land tenure issues or community conflicts
- Environmental regulations affecting operations
- Infrastructure development delays

Climate Risks:

- Drought or excessive rainfall affecting yields
- Climate change impacts on optimal growing conditions
- Increased frequency of extreme weather events

11. RECOMMENDATIONS

11.1 Strategic Recommendation

Based on the comprehensive analysis, we recommend pursuing a **phased development approach targeting the Optimistic scenario in Cross River State**. This strategy balances risk management with the potential for strong returns. **Phase 1: Pilot Project (Years 1-3)**

• Establish 500-hectare pilot plantation in Cross River State
• Investment requirement: ■1.5-2.0 billion
• Objectives:
- Validate yield and cost assumptions
- Build operational capabilities and local partnerships
- Secure additional land options for expansion
- Establish processing infrastructure

Phase 2: Scale-Up (Years 4-7)

• Expand to full 1,339 hectares based on pilot performance
• Additional investment: ■2.8-3.0 billion
• Objectives:
- Achieve target production levels
- Reach ■1 billion annual profit target
- Consider geographic diversification to Ondo State

11.2 Critical Success Factors

• Achieve and maintain oil extraction rate of 21% or higher
• Sustain FFB yields of 18-21 MT per hectare through best practices
• Secure competitive long-term pricing through forward contracts or vertical integration
• Invest in modern, efficient processing facilities
• Build strong community relationships and local partnerships
• Implement rigorous financial and operational monitoring systems

11.3 Next Steps

Immediate (0-3 months):

1. Commission detailed soil surveys of potential land in Cross River
2. Engage with palm oil processors for off-take agreements
3. Develop detailed financial model with sensitivity analysis
4. Initiate legal review of land acquisition requirements
5. Identify potential joint venture partners with local expertise

Near-term (3-6 months):

6. Complete environmental impact assessment
7. Secure land options or preliminary agreements
8. Finalize processing facility design and cost estimates
9. Develop detailed operational plan and management structure
10. Secure financing commitments

11.4 Data Gaps to Address

Before final investment decision, obtain:

• Detailed soil analysis reports for specific land parcels
• Updated market price forecasts and volatility analysis
• Comprehensive competitive landscape assessment
• Firm cost quotes from equipment suppliers and contractors
• Environmental and social impact assessment results
• Community consultation outcomes

CONCLUSION

This technical analysis demonstrates that achieving ■1 billion in annual profit from palm oil production in Nigeria is feasible but requires careful planning, significant investment, and operational excellence. The study identifies Cross River State as the optimal location, offering the best combination of agro-climatic conditions, infrastructure, and profitability potential. The recommended phased approach—beginning with a 500-hectare pilot and scaling to 1,339 hectares—provides a prudent path forward. Under the Optimistic scenario, this strategy offers a 22.8% ROI with a 7.4-year payback period, representing an attractive investment opportunity. However, success depends on multiple factors including achieving competitive yields, maintaining efficient operations, and navigating market volatility. The Conservative scenario's unprofitability underscores the importance of targeting best-in-class performance. This feasibility study provides a robust analytical foundation for the investment decision. However, it should be supplemented with detailed site-specific assessments, market analysis, and risk management strategies before final commitment of capital. The palm oil sector in Nigeria presents significant opportunities for investors who can achieve operational excellence and scale. With proper execution, this investment has the potential to deliver both strong financial returns and positive economic impact in the target regions.